

Homework 2

1.4.6 Let f be a measurable function on $(\mathbb{R}^n, \lambda)$. Define the *centered maximal function of f with respect to cubes*, by

$$\mathcal{M}_c(f)(x) = \sup_{\varepsilon > 0} \frac{1}{|Q(x, \varepsilon)|} \int_{Q(x, \varepsilon)} |f(y)| dy.$$

Where $Q(x, \varepsilon)$ is a cube parallel to the axes with side length 2ε centered at x .

a. Prove that $\mathcal{M}_c(f) \approx \mathcal{M}(f)$.

Proof. Consider the n -cubes with side length 2ε and the n -balls with radius ε . We have that $B(x, \varepsilon) \subseteq Q(x, \varepsilon)$. Let $\alpha = |B(x, \varepsilon)|/|Q(x, \varepsilon)|$.

$$\begin{aligned} \int_{B(x, \varepsilon)} |f| d\lambda &\leq \int_{Q(x, \varepsilon)} |f| d\lambda. \\ \frac{1}{|B(x, \varepsilon)|} \cdot \frac{1}{|Q(x, \varepsilon)|} \int_{B(x, \varepsilon)} |f| d\lambda &\leq \frac{1}{|B(x, \varepsilon)|} \cdot \frac{1}{|Q(x, \varepsilon)|} \int_{Q(x, \varepsilon)} |f| d\lambda. \\ \frac{1}{|B(x, \varepsilon)|} \int_{B(x, \varepsilon)} |f| d\lambda &\leq \frac{|Q(x, \varepsilon)|}{|B(x, \varepsilon)|} \cdot \frac{1}{|Q(x, \varepsilon)|} \int_{Q(x, \varepsilon)} |f| d\lambda. \\ \alpha \cdot \frac{1}{|B(x, \varepsilon)|} \int_{B(x, \varepsilon)} |f| d\lambda &\leq \frac{1}{|Q(x, \varepsilon)|} \int_{Q(x, \varepsilon)} |f| d\lambda. \\ \alpha \cdot \sup_{\varepsilon > 0} \left\{ \frac{1}{|B(x, \varepsilon)|} \int_{B(x, \varepsilon)} |f| d\lambda \right\} &\leq \sup_{\varepsilon > 0} \left\{ \frac{1}{|Q(x, \varepsilon)|} \int_{Q(x, \varepsilon)} |f| d\lambda \right\}. \\ \alpha \cdot \mathcal{M}(f) &\leq \mathcal{M}_c(f). \end{aligned}$$

Now consider the n -balls of radius $\sqrt{n}\varepsilon$. Let $\beta = |B(x, \sqrt{n}\varepsilon)|/|Q(x, \varepsilon)|$, which is clearly constant. Then

$$\begin{aligned} \int_{B(x, \sqrt{n}\varepsilon)} |f| d\lambda &\geq \int_{Q(x, \varepsilon)} |f| d\lambda. \\ \frac{1}{|B(x, \sqrt{n}\varepsilon)|} \cdot \frac{1}{|Q(x, \varepsilon)|} \int_{B(x, \sqrt{n}\varepsilon)} |f| d\lambda &\geq \frac{1}{|B(x, \sqrt{n}\varepsilon)|} \cdot \frac{1}{|Q(x, \varepsilon)|} \int_{Q(x, \varepsilon)} |f| d\lambda. \\ \frac{1}{|B(x, \sqrt{n}\varepsilon)|} \int_{B(x, \sqrt{n}\varepsilon)} |f| d\lambda &\geq \frac{|Q(x, \varepsilon)|}{|B(x, \sqrt{n}\varepsilon)|} \cdot \frac{1}{|Q(x, \varepsilon)|} \int_{Q(x, \varepsilon)} |f| d\lambda. \\ \beta \frac{1}{|B(x, \sqrt{n}\varepsilon)|} \int_{B(x, \sqrt{n}\varepsilon)} |f| d\lambda &\geq \frac{1}{|Q(x, \varepsilon)|} \int_{Q(x, \varepsilon)} |f| d\lambda. \\ \beta \mathcal{M}(f) &\geq \mathcal{M}_c(f). \end{aligned}$$

We have shown that

$$\alpha \cdot \mathcal{M}(f) \leq \mathcal{M}_c(f) \leq \beta \mathcal{M}(f).$$

Which is equivalent to $\mathcal{M}(f) \approx \mathcal{M}_c(f)$. □

b. Prove that $\mathcal{M}_c(f) \approx M_c(f)$.

Proof. The fact that $\mathcal{M}_c(f) \leq M_c(f)$ is obvious. We now must show that $M_c(f) \leq \delta \mathcal{M}_c(f)$. Define $d' : \mathbb{R}^n \times \mathbb{R}^n \rightarrow [0, \infty)$ by $d' : (x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n) \mapsto \max\{|x_1 - y_1|, |x_2 - y_2|, \dots, |x_n - y_n|\}$. let $x_0 \in \{y : d'(x, y) \leq \frac{\varepsilon}{2}\} = Y$. Then $Q(x_0, \frac{\varepsilon}{2}) \subseteq Q(x, \varepsilon)$. Define $|Q(x_0, \frac{\varepsilon}{2})|/|Q(x, \varepsilon)| = |Q(x, \frac{\varepsilon}{2})|/|Q(x, \varepsilon)| = \alpha$.

$$\begin{aligned} \int_{Q(x_0, \frac{\varepsilon}{2})} |f| d\lambda &\leq \int_{Q(x, \varepsilon)} |f| d\lambda. \\ \frac{1}{|Q(x_0, \frac{\varepsilon}{2})|} \cdot \frac{1}{|Q(x, \varepsilon)|} \int_{Q(x_0, \frac{\varepsilon}{2})} |f| d\lambda &\leq \frac{1}{|Q(x_0, \frac{\varepsilon}{2})|} \cdot \frac{1}{|Q(x, \varepsilon)|} \int_{Q(x, \varepsilon)} |f| d\lambda. \\ \frac{1}{|Q(x_0, \frac{\varepsilon}{2})|} \int_{Q(x_0, \frac{\varepsilon}{2})} |f| d\lambda &\leq \frac{|Q(x, \varepsilon)|}{|Q(x_0, \frac{\varepsilon}{2})|} \cdot \frac{1}{|Q(x, \varepsilon)|} \int_{Q(x, \varepsilon)} |f| d\lambda. \\ \alpha \cdot \frac{1}{|Q(x_0, \frac{\varepsilon}{2})|} \int_{Q(x_0, \frac{\varepsilon}{2})} |f| d\lambda &\leq \frac{1}{|Q(x, \varepsilon)|} \int_{Q(x, \varepsilon)} |f| d\lambda. \\ \alpha \cdot \sup_{\varepsilon > 0} \left\{ \frac{1}{|Q(x_0, \frac{\varepsilon}{2})|} \int_{Q(x_0, \frac{\varepsilon}{2})} |f| d\lambda \right\} &\leq \sup_{\varepsilon > 0} \left\{ \frac{1}{|Q(x, \varepsilon)|} \int_{Q(x, \varepsilon)} |f| d\lambda \right\}. \\ \alpha \cdot M_c(f) &\leq \mathcal{M}_c(f). \end{aligned}$$

So we have that

$$\mathcal{M}_c(f) \leq M_c(f) \leq \alpha^{-1} \mathcal{M}_c(f).$$

Which is equivalent to $\mathcal{M}_c(f) \approx M_c(f)$. □

c. Show that for any cube Q with side length $l(Q)$ we have

$$M_c(\chi_Q)(x) \approx \frac{l(Q)^n}{(l(Q) + d_x)^n}.$$

Proof. We wish to show that $M_c(\chi_Q) \approx \frac{l(Q)^n}{(l(Q) + d_x)^n}$. First, suppose the $d_x > 0$.

$$M_c(\chi_Q) \cdot \frac{(l(Q) + d_x)^n}{l(Q)^n} \geq \frac{l(Q)^n}{(l(Q) + d_x)^n} \cdot \frac{(l(Q) + d_x)^n}{l(Q)^n} = 1.$$

This is by the definition of M_c involving a supremum and a standard choice of n-cube. We now show boundedness from above. We showed in the previous parts that $M_c(f) \approx \mathcal{M}(f)$. So

$$M_c(\chi_Q) \cdot \frac{(l(Q) + d_x)^n}{l(Q)^n} \leq \rho \sup_{\varepsilon > 0} \left\{ \frac{1}{|B(x, \varepsilon)|} \int_{B(x, \varepsilon)} \chi_Q d\lambda \right\} \cdot \frac{(l(Q) + d_x)^n}{l(Q)^n}.$$

Notice that this supremum must be when $d_x + l(Q) \leq \varepsilon \leq d_x + \sqrt{n}l(Q)$. Thus, we have that

$$\rho \sup_{\varepsilon > 0} \left\{ \frac{1}{|B(x, \varepsilon)|} \int_{B(x, \varepsilon)} \chi_Q d\lambda \right\} \cdot \frac{(l(Q) + d_x)^n}{l(Q)^n} \leq \frac{\rho l(Q)^n}{v_n (d_x + l(Q))^n} \cdot \frac{(l(Q) + d_x)^n}{l(Q)^n} = \frac{\rho}{v_n}.$$

Now, suppose $d_x = 0$. $M_c(\chi_Q) = 1$ because we can pick an appropriately placed, cube with side length smaller than $l(Q)$ that is entirely contained in Q . Thus, $M_c(\chi_Q) = \frac{(l(Q)+d_x)^n}{l(Q)^n} = 1$.

So, $\max\{1, \rho/v_n\}$ is an upper bound. Thus, we have shown that $M_c(\chi_Q) \approx l(Q)^n/(l(Q) + d_x)^n$. $\square$

- d. Conclude that for any concentric multiple Q^* of Q we have $M_c(\chi_Q) \approx M_c(\chi_{Q^*})$.

Proof. Let Q^* be a concentric multiple of Q with $l(Q^*) = tl(Q)$. Additionally, without loss of generality, assume Q is centered at 0 and $t > 1$.

We begin by considering the set $C_s(Q) = \bigcup_{i \leq n} \{(x_1, x_2, \dots, x_i, \dots, x_n) \in \mathbb{R}^n : |x_k| \leq l(Q)/2, k \leq n, k \neq i\}$. This is essentially the subset of $\mathbb{R}^n$ that is the projection of Q along all coordinate axes.

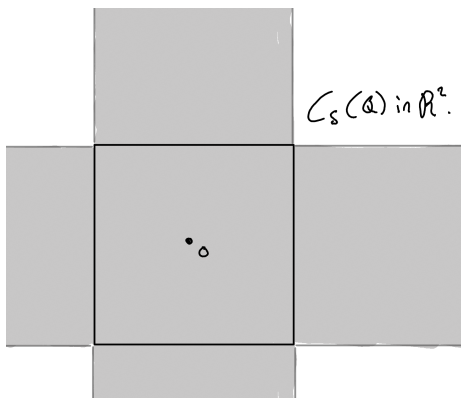


Figure 1: C_s in $\mathbb{R}^2$

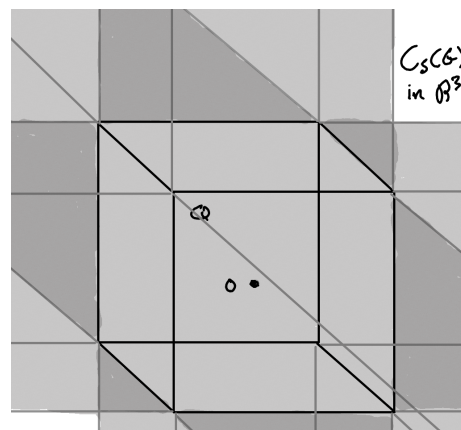


Figure 2: C_s in $\mathbb{R}^3$

let $x \in C_s(Q) \setminus Q^*$. Define d_{x^*} to be the distance from x to Q^* . Then

$$d_x^* = d_x - \frac{l(Q^*) - l(Q)}{2} = d_x - \frac{(t-1)l(Q)}{2}.$$

so,

$$\begin{aligned} \frac{l(Q)}{l(Q) + d_x} \cdot \frac{d_x^* + l(Q^*)}{l(Q^*)} &= \frac{l(Q)}{l(Q) + d_x} \cdot \frac{d_x - \frac{(t-1)l(Q)}{2} + tl(Q)}{tl(Q)} \\ &= \frac{d_x - \frac{(t-1)l(Q)}{2} + tl(Q)}{tl(Q) + td_x} := \tau. \end{aligned}$$

Let us now define $\xi(\chi_Q) = \frac{l(Q)}{l(Q)+d_x}$. Then, by part (c), $\xi(\chi_Q)^n \approx M_c(\chi_Q)$. Notice that $\tau = \xi(\chi_Q)/\xi(\chi_{Q^*})$. Clearly, $\frac{1}{t} \leq \tau = \xi(\chi_Q)/\xi(\chi_{Q^*})$. Thus:

$$\frac{1}{t^n} \leq \frac{\xi(\chi_Q)^n}{\xi(\chi_{Q^*})^n}.$$

Furthermore, by (c), we have that $\xi(\chi_Q)^n \leq \alpha M_c(\chi_Q)$ and $\xi(\chi_{Q^*})^{-n} \leq (\beta M_c(\chi_{Q^*}))^{-1}$. Putting this all together, we have that:

$$\frac{1}{t^n} \leq \frac{\xi(\chi_Q)^n}{\xi(\chi_{Q^*})^n} \leq \frac{\alpha}{\beta} \cdot \frac{M_c(\chi_Q)}{M_c(\chi_{Q^*})} \implies \frac{\beta}{\alpha t^n} \leq \frac{M_c(\chi_Q)}{M_c(\chi_{Q^*})} \leq 1.$$

Thus, $M_c(\chi_Q) \approx M_c(\chi_{Q^*})$ when $x \in C_s(Q) \setminus Q^*$.

Now consider $x \in \mathbb{R}^n \setminus (C_s(Q) \cup Q^*)$. Then $d_{x^*} \geq d_x - \frac{\sqrt{n}}{2}(t-1)l(Q)$. So $\frac{tl(Q)}{tl(Q)+d_{x^*}} \leq \frac{tl(Q)}{tl(Q)+d_x - \frac{\sqrt{n}}{2}(t-1)l(Q)}$. Which gives:

$$\begin{aligned} \frac{l(Q)}{l(Q)+d_x} \cdot \frac{tl(Q)+d_{x^*}}{tl(Q)} &\geq \frac{l(Q)}{l(Q)+d_x} \cdot \frac{tl(Q)+d_x - \frac{\sqrt{n}}{2}(t-1)l(Q)}{tl(Q)} \\ &= \frac{tl(Q)+d_x - \frac{\sqrt{n}}{2}(t-1)l(Q)}{tl(Q)+td_x} := \tau. \end{aligned}$$

Notice that this fraction can be negative, thus for small d_x we will need a better estimate. First, consider the case when $d_x > \frac{\sqrt{n}}{2}(t-1)l(Q)$. We get:

$$\tau \geq \frac{tl(Q)}{tl(Q)+t\frac{\sqrt{n}}{2}(t-1)l(Q)} = \frac{t}{t+t\frac{\sqrt{n}}{2}(t-1)} = \frac{1}{1+\frac{\sqrt{n}}{2}(t-1)}.$$

So, $\tau \geq \frac{1}{1+\frac{\sqrt{n}}{2}(t-1)}$. Therefore we have

$$\left(\frac{1}{1+\frac{\sqrt{n}}{2}(t-1)}\right)^n \leq \left(\frac{\xi(\chi_Q)}{\xi(\chi_{Q^*})}\right)^n \leq \frac{\alpha}{\beta} \cdot \frac{M_c(\chi_Q)}{M_c(\chi_{Q^*})}.$$

Therefore

$$\frac{\beta}{\alpha} \left(\frac{1}{1+\frac{\sqrt{n}}{2}(t-1)}\right)^n \leq \frac{M_c(\chi_Q)}{M_c(\chi_{Q^*})} \leq 1.$$

So $M_c(\chi_Q) \approx M_c(\chi_{Q^*})$.

Now consider the case where $d_x \leq \frac{\sqrt{n}}{2}(t-1)l(Q)$. We apply part (a). Therefore $M(\chi_Q) \approx M_c(\chi_Q)$. Let $\delta = \frac{\sqrt{n}}{2}tl(Q)$.

$$\begin{aligned} M_c(\chi_Q) &\geq \kappa \sup_{\varepsilon>0} \frac{1}{|B(x,\varepsilon)|} \int_{B(x,\varepsilon)} \chi_Q d\lambda, \text{ where } \lambda \text{ denotes Lebesgue measure.} \\ &\geq \kappa \frac{1}{|B(x,\delta)|} \int_{B(x,\delta)} \chi_Q d\lambda. \\ &= \frac{\kappa l(Q)^n}{v_n \delta^n}. \end{aligned}$$

And clearly, $M_c(\chi_{Q^*}) \leq 2 \implies \frac{1}{2} \leq M_c(\chi_{Q^*})^{-1}$. So

$$1 \geq \frac{M_c(\chi_Q)}{M_c(\chi_{Q^*})} \geq \frac{\kappa l(Q)^n}{2v_n \delta^n}.$$

so, we have shown that $M_c(\chi_Q) \approx M_c(\chi_{Q^*})$.

If $x \in Q$, then $M_c(\chi_Q) = M_c(\chi_{Q^*})$.

Finally, if $x \in Q^* \setminus Q$, then $M_c(\chi_Q)/M_c(\chi_{Q^*}) = M_c(\chi_Q)$.

$$d_x \leq (l(Q^*) - l(Q)) \frac{\sqrt{n}}{2} = \frac{(t-1)l(Q)\sqrt{n}}{2}.$$

Then $\frac{l(Q)}{l(Q)+d_x} \geq \frac{l(Q)}{l(Q) + \frac{(t-1)l(Q)\sqrt{n}}{2}} = \frac{1}{1 + \frac{(t-1)\sqrt{n}}{2}}.$

So, $\alpha \cdot \frac{1}{1 + \frac{(t-1)\sqrt{n}}{2}} \leq M_c(\chi_Q) \leq 1.$

Therefore $M_c(\chi_Q) \approx M_c(\chi_{Q^*})$. We are done. □